

CHEM-1310
Fall 2017
Third Examination
Form A

Multiple Choice - Choose the BEST Answer

Draw the Lewis structure of BrF_5 and use it to answer the next four questions.

1. How many regions of electron density are present on the central atom in BrF_5 ?

- A) 2
- B) 3
- C) 4
- D) 5
- E) 6

2. What is the electron geometry of BrF_5 ?

- A) square planar
- B) octahedral
- C) trigonal bipyramidal
- D) seesaw
- E) tetrahedral

3. What is the molecular geometry of BrF_5 ?

- A) square pyramidal
- B) trigonal pyramidal
- C) trigonal bipyramidal
- D) seesaw
- E) t-shaped

4. The molecular BrF_5 contains which of the following

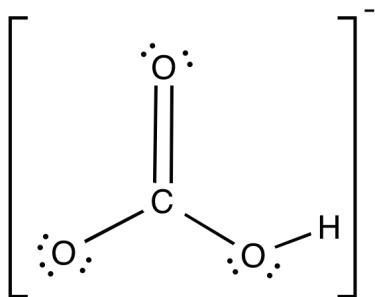
- A) ionic bonds and is a polar molecule
- B) non-polar covalent bonds and is a nonpolar molecule
- C) polar covalent bonds and is a polar molecule
- D) nonpolar covalent bonds and is a polar molecule
- E) polar covalent bonds and is a nonpolar molecule

Multiple Choice - Choose the BEST Answer

5. The basis of the VSEPR model of molecular bonding is _____.

- A) regions of electron density in the valence shell of an atom will arrange themselves to minimize repulsions
- B) hybrid orbitals will form as necessary to, as closely as possible, achieve spherical symmetry
- C) regions of electron density in the valence shell of an atom will arrange themselves to maximize overlap
- D) regions of electron density on an atom will organize themselves so as to maximize s-character
- E) atomic orbitals of the bonding atoms become hybrid orbitals.

6. Which of the following statements is true regarding the ion shown below?

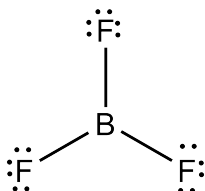


- I. The ion has a resonance structure.
- II. The O-H bond in the ion is the strongest bond present.
- III. There are exactly three σ bonds and one π bond in the ion.

- A) I and II only
- B) I and III only
- C) II only
- D) I only
- E) I, II and III

Multiple Choice - Choose the BEST Answer

7. According to valence bond theory, which orbitals overlap in the formation of the σ B–F bond in the molecular below?

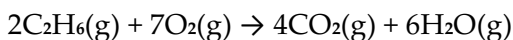


- A) 2p on B and 2p on F
 - B) **sp^2 on B and 2p on F**
 - C) sp^3 on B and sp^3 on F
 - D) sp on B and sp^3 on F
 - E) The Lewis structure shown above is incorrect; therefore, the orbital formation of the B-F bond cannot be determined.
8. Which of the following correctly explains why we use hybrid orbitals to describe bonding in CH_4 ?
- A) **Carbon's s and p orbitals have different energies and are not spatially equivalent.**
 - B) Electrons need to be promoted to a higher energy subshell.
 - C) Carbon's orbitals need to be equivalent to hydrogen's orbitals to produce the greatest overlap.
 - D) Hydrogen's s orbitals are too high in energy to form a bond with carbon's p orbitals.
 - E) None of the above are true statements explaining why we use hybrid orbitals to describe bonding in CH_4 .
9. You have a pure sample of $CH_3OH(l)$, which of the following lists all of the intermolecular interactions that occur in the sample.

- A) hydrogen bonding, ion-dipole and London dispersion interactions
- B) hydrogen bonding and dipole-dipole interactions
- C) **hydrogen bonding, dipole-dipole and London dispersion interactions**
- D) dipole-dipole, ion-dipole and London dispersion interactions
- E) dipole-dipole and London dispersion interactions

Multiple Choice - Choose the BEST Answer

10. What is the enthalpy of combustion for ethane, C_2H_6 , as described by the following equation, in kJ per mol of reaction?



Use the information below as needed.

Average Bond Dissociation Energies (kJ/mol)

H-C	414
H-O	464
C-O	360
C=O	736*
O=O	498
C-C	347
C=C	611

- A) -2.31×10^3
- B) -6.65×10^2
- C) 3.51×10^2
- D) 288
- E) There is not enough information.

11. When a soluble ionic compound is placed in water, which of the following occurs?

- A) The ionic compound dissociates into its ions
- B) Hydrogen bonding occurs between the water molecules present
- C) Ion-dipole interaction occurs between the water molecules and the ions present.
- D) Hydration occurs
- E) All of the above occur.

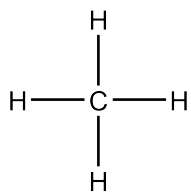
12. Which of the following molecules have the highest vapor pressure?

- A) NH_3
- B) H_2O
- C) CH_4
- D) HCl
- E) CH_2Cl_2

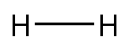
Multiple Choice - Choose the BEST Answer

13. Which of the following molecules is the most polarizable?

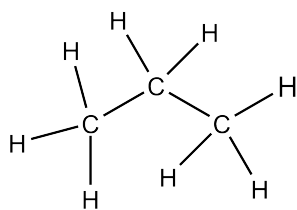
A)



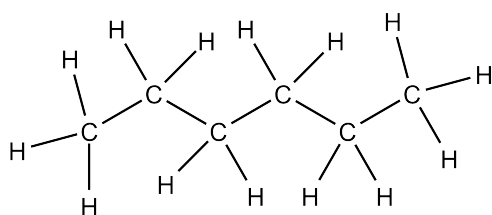
B)



C)



D)



E) All are equally polarizable.

Multiple Choice - Choose the BEST Answer

14. Which process releases the greatest amount of heat to the surroundings?

$$\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$$

$$\Delta H_{\text{fus}} = 6.02 \text{ kJ/mol}$$

- A) the condensation of 10.0 g of gaseous H₂O
- B) the freezing of 10.0 g of liquid H₂O
- C) the boiling of 10.0 g of liquid H₂O
- D) the melting of 10.0 g of solid H₂O
- E) the sublimation of 10.0 g of solid H₂O

15. How much energy (in kJ) is required to completely condense 200.0 g of 125.00°C steam?

Use the information below as needed.

Properties of Water

Specific Heats:

$$\text{gas} = 1.84 \text{ J/g}^\circ\text{C}$$

$$\text{liquid} = 4.184 \text{ J/g}^\circ\text{C}$$

$$\text{solid} = 2.09 \text{ J/g}^\circ\text{C}$$

$$\text{Enthalpy of Vaporization: } \Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$$

$$\text{Enthalpy of fusion: } \Delta H_{\text{fus}} = 6.02 \text{ kJ/mol}$$

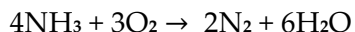
- A) -452 kJ
- B) -406 kJ
- C) -461 kJ
- D) +20.92 kJ
- E) +46 kJ

16. A first-order reaction is 38.5% complete in 520 s. What is the value of the rate constant?

- A) $1.83 \times 10^{-3} \text{ s}^{-1}$
- B) $9.35 \times 10^{-4} \text{ s}^{-1}$
- C) $3.07 \times 10^{-3} \text{ s}^{-1}$
- D) $1.18 \times 10^{-3} \text{ s}^{-1}$
- E) $1.20 \times 10^{-3} \text{ s}^{-1}$

Multiple Choice - Choose the BEST Answer

17. For the following reaction it was found that the rate of formation of N_2 was 0.27 M/s .



At what rate was ammonia being consumed?

- A) -0.27 M/s
- B) 0.54 M/s
- C) 0.81 M/s
- D) 0.13 M/s
- E) 0.77 M/s

18. For the reaction:



a set of experiments are performed, with different initial conditions at the same temperature the data is shown below. What is the rate law for the reaction?

Experiment	$[\text{BrO}_3^-] \text{ (M)}$	$[\text{Br}^-] \text{ (M)}$	$[\text{H}^+] \text{ (M)}$	Initial Rate (M/s)
1	0.10	0.10	0.10	8.0×10^{-4}
2	0.20	0.10	0.10	1.6×10^{-3}
3	0.20	0.20	0.10	3.2×10^{-3}
4	0.10	0.10	0.20	3.2×10^{-3}

- A) $\text{Rate} = k[\text{Br}^-][\text{H}^+]^2$
- B) $\text{Rate} = k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]$
- C) $\text{Rate} = k[\text{BrO}_3^-][\text{H}^+]$
- D) $\text{Rate} = k[\text{H}^+]$
- E) $\text{Rate} = k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$

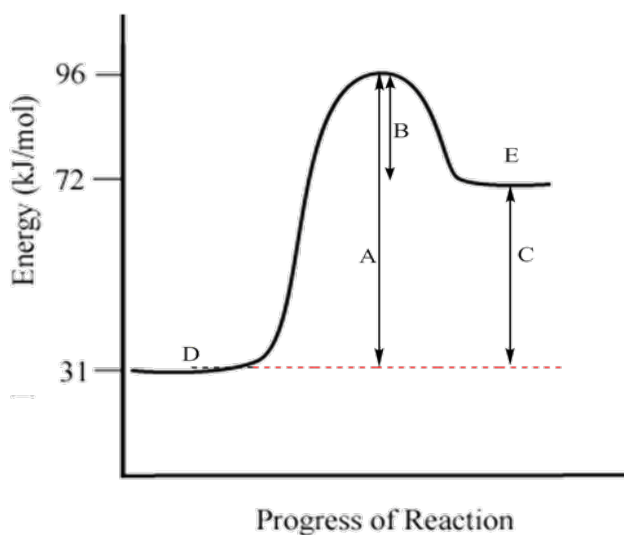
Multiple Choice - Choose the BEST Answer

19. For two molecules to react when they collide, which of the following must occur?

- I. proper orientation of molecules
- II. exothermic reaction
- III. sufficient kinetic energy

- A) I and II only
- B) I and III only
- C) II only
- D) I only
- E) I, II and III

20. Which of the following is not a true statement about the graph below?



- A) The reaction is endothermic.
- B) Line 'C' represents the enthalpy of reaction.
- C) Line 'A' represents the activation energy of the reaction
- D) The transition state of the reaction occurs at the top of the potential energy curve.
- E) All of the above statements are correct.

21. Which form of the exam do you have?

- A) Form A
- B) Form B